Derivation of an Intuitive Imputation Formula for Missing LVEF

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Estimating Missing Ejection Fraction

Left Ventricular Ejection Fraction (LVEF) is simply the fraction of blood pumped out of the ventricle per contraction. By definition, it is stroke volume (SV) divided by total end-diastolic filling volume (EDV):

$$\text{LVEF} = \frac{SV}{EDV}$$

The total volume pumped out per minute is Cardiac Output (CO), which combines heart rate (HR) with stroke volume:

$$CO = HR \times SV = HR \times EDV \times \text{LVEF}$$

From basic fluid dynamics, pressure roughly equals flow multiplied by peripheral resistance (R). Approximating systolic pressure (SBP) from forward flow gives:

$$SBP \approx CO \times R = (HR \times EDV \times \text{LVEF}) \times R$$

Rearranging this relationship for pumping capacity shows that LVEF varies directly with blood pressure and inversely with heart rate:

$$\text{LVEF} \propto \frac{SBP}{HR}$$

In emergency medicine, the ratio of heart rate over systolic blood pressure ($\frac{HR}{SBP}$) is called the Shock Index. When the heart muscle weakens and ejection fraction falls, blood pressure drops and the body increases heart rate to keep up. Because of this inverse relationship:

$$\text{LVEF} \propto \frac{1}{\text{Shock Index}}$$

We approximate this loss in pumping ability linearly as $-\beta\left(\frac{HR}{SBP}\right)$, where β scales the bedside shock ratio.

Accounting for Lung Fluid and Heart Attack Severity

Beyond vital signs, two clinical factors directly reduce ejection fraction in acute coronary syndrome. First, when the left ventricle fails, fluid backs up into the lungs. This congestion is graded clinically by Killip class from 1 (clear lungs) to 4 (cardiogenic shock).

Because each grade increase reflects deeper failure, we deduct a fixed amount α for every step above class 1, written as $-\alpha(K-1)$.

Second, patients with a full-thickness ST-elevation myocardial infarction (STEMI) suffer complete vessel blockage and greater muscle necrosis compared to non-STEMI or unstable angina presentations. We capture this sudden wall injury using an indicator variable $\mathbb{I}_{\text{STEMI}}$ (equal to 1 for STEMI and 0 otherwise), applying an additive penalty $-\gamma\mathbb{I}_{\text{STEMI}}$.

Combined Formula and Dataset Calibration

Combining these physiological deductions from an uncomplicated baseline (LVEF_0), our working estimation formula becomes:

$$\widehat{\text{LVEF}} = \text{LVEF}_0 - \alpha(K-1) - \beta\left(\frac{HR}{SBP}\right) - \gamma\mathbb{I}_{\text{STEMI}}$$

To ensure values remain biologically realistic, the final estimate is bounded within physiological limits:

$$\widehat{\text{LVEF}}_{\text{final}} = \min\left(70,\, \max\left(15,\, \widehat{\text{LVEF}}\right)\right)$$

From our hospital cohort (n = 1,054 complete records), the baseline average for uncomplicated non-STEMI cases is $\text{LVEF}_0 = 52.0\%$. Patients presenting with STEMI showed an average drop of $\gamma = 5.1\%$, while each Killip stage escalation accounted for an average decrease of $\alpha = 3.5\%$. Setting the shock index weight to $\beta = 4.0$ matches the observed clinical range, yielding our applied formula:

$$\widehat{\text{LVEF}} = 52.0 - 3.5(K-1) - 4.0\left(\frac{HR}{SBP}\right) - 5.1\mathbb{I}_{\text{STEMI}}$$